

app

Fraud analytics

Fraud Prediction

Model performance

Project info

Risk Monitoring



Financial Fraud Detection & Risk Monitoring System

Project Overview

This project detects fraudulent financial transactions using Machine Learning and Analytics.

Features

- Fraud Analytics
- ML Model Performance
- Fraud Prediction
- Risk Monitoring
- Business Insights

Dashboard Loaded Successfully

app

Fraud analytics

Fraud Prediction

Model performance

Project info

Risk Monitoring

 Fraud Analytics

Transactions

50000

Fraud Cases

16067

Fraud Rate

32.13

Fraud Distribution

app

Fraud analytics

Fraud Prediction

Model performance

Project info

Risk Monitoring



Fraud Prediction

Transaction Amount

0.00

—

+

Previous Fraud Activity

0

▼

Predict

- app
- Fraud analytics
- Fraud Prediction
- Model performance
- Project info
- Risk Monitoring

Model Performance

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression	0.92	0.91	0.89	0.9
1	Random Forest	0.97	0.96	0.95	0.95
2	XGBoost	0.98	0.97	0.96	0.96

Best Model : XGBoost



Confusion Matrix



app

Fraud analytics

Fraud Prediction

Model performance

Project info

Risk Monitoring



Project Information

Tech Stack

- Python
- Pandas
- Scikit Learn
- XGBoost
- Streamlit

Dataset

- Synthetic Fraud Dataset
- Credit Card Fraud Dataset

Objective

Detect fraudulent transactions using machine learning.

Risk Monitoring



	Transaction_ID	User_ID	Transaction_Amount	Transaction_Type	Date	Account_Balance	Device_Type	Location	Merchant_Category	Previous_Fraudulent_Activity	Daily_Transaction_Count	Card_Type	Card_Age	Fraud_Label
1	TXN_9427	USER_7875	1.19	Bank Transfer	7 June 2023	75725.25	Mobile	New York	Clothing	0	13	Mastercard	186	1
2	TXN_199	USER_2734	28.96	Online	20 June 2023	1588.96	Tablet	Mumbai	Restaurants	0	14	Visa	226	1
3	TXN_12447	USER_2617	254.32	ATM Withdrawal	7 December 2023	76807.2	Tablet	New York	Clothing	0	8	Visa	76	1
4	TXN_39489	USER_2014	31.28	POS	11 November 2023	92354.66	Mobile	Mumbai	Electronics	1	14	Mastercard	140	1
7	TXN_49498	USER_4660	7.08	ATM Withdrawal	25 February 2023	45826.27	Tablet	London	Restaurants	0	3	Discover	182	1
12	TXN_6188	USER_6728	55.5	ATM Withdrawal	20 July 2023	89664.63	Laptop	London	Groceries	0	6	Discover	192	1
18	TXN_12609	USER_5861	178.56	POS	16 August 2023	62359.52	Tablet	Tokyo	Groceries	0	4	Visa	29	1
21	TXN_15908	USER_4615	203.97	ATM Withdrawal	18 June 2023	73616.91	Mobile	New York	Restaurants	0	4	Mastercard	173	1
24	TXN_13466	USER_9775	72.33	ATM Withdrawal	14 August 2023	29954.64	Mobile	Sydney	Clothing	0	3	Discover	120	1
28	TXN_37237	USER_1116	8.06	ATM Withdrawal	9 July 2023	38155.13	Laptop	Sydney	Travel	0	9	Mastercard	232	1